WeanAge user guide

Estimating weaning age from $\delta^{15}N$ in tooth dentine and bone collagen

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About this guide
1. Background
- Why $\delta^{15}N$ records weaning
- What “weaning age” means here
- The routes through the software

2. Data and quality control
3. Turning tooth position into age
4. The tooth calculation
- Curve models
- When no age is reported
- Uncertainty
- Choosing and averaging models
- Corrections that can be switched on
- Warnings

5. The bone calculation
6. Season of birth and rumen development (enamel)
7. Comparing groups
8. Planning a study
9. Using the app
10. Limits and placeholders
References

About this guide

WeanAge estimates when young mammals were weaned from the nitrogen isotope ratio ($\delta^{15}N$) of their collagen. It works with two kinds of material: thin slices along one tooth, which record one animal’s childhood, and the bones of many young animals from one site that died at different ages. For cattle it also reads enamel $\delta^{18}O$ and $\delta^{13}C$ along a molar for the season of birth and the development of the rumen. It runs as a local app or in a web browser.

This guide explains each part of the software, how the parts combine into a weaning age, and where each method comes from. Section 1 gives the background, sections 2–8 follow the calculations step by step, section 9 shows the app, and section 10 lists what is still provisional. Words in `code font` are column names, settings or files in the software. References are listed at the end.

1. Background

Why $\delta^{15}N$ records weaning

$\delta^{15}N$ rises by about 3–4‰ with each step up a food chain (DeNiro & Epstein 1981; Minagawa & Wada 1984). A suckling animal feeds on its mother’s tissues through her milk, so its own collagen is raised above hers, usually by 2–3‰ in practice (Balasse et al. 2001; Fuller et al. 2006). When milk is replaced by solid food, new collagen gradually falls to the adult level. The age at which that fall happens is the weaning age.

Collagen is the protein measured. Dentine collagen forms in layers as the tooth grows and is not renewed afterwards, so a series of slices from crown tip to root is a timeline of one animal’s diet (Beaumont et al. 2013). Bone collagen, by contrast, is renewed throughout life, so a bone records a mix of recent and older diet; a single bone cannot give a timeline, but many young animals that died at different ages can (Tsutaya & Yoneda 2013).

What “weaning age” means here

Weaning is a process, not an instant. WeanAge reports three milestones of the $\delta^{15}N$ fall (Figure 1): **onset** (5% of the fall has happened), **midpoint** (half of it) and **completion** (95%). The midpoint is reported by default: the curve is steepest there, so it is dated most precisely. Completion depends on how slowly $\delta^{15}N$ levels off and can fall well after suckling stopped. In known-age calves the $\delta^{15}N$ fall traces the end of suckling (Balasse et al. 2001), and the same has been found in sheep (Díez-Canseco & Tornero 2026).

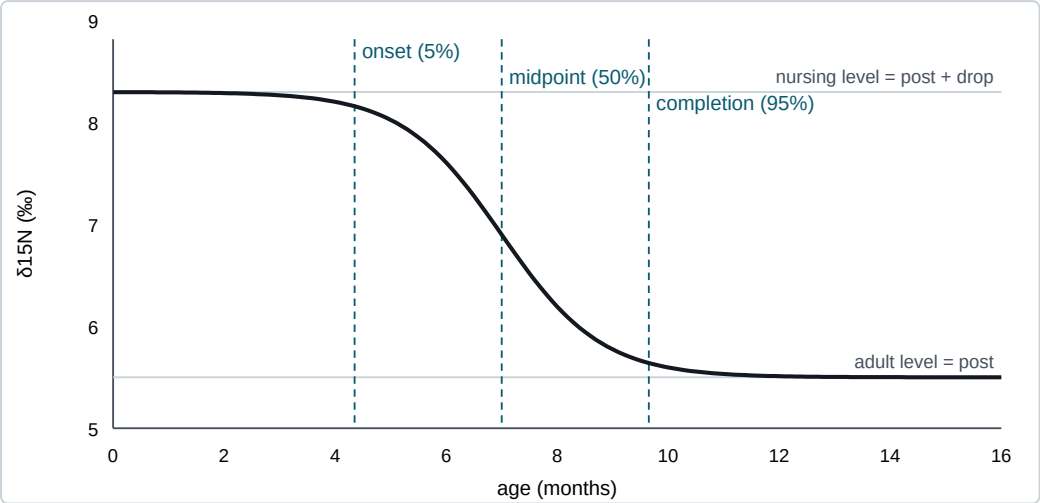


Figure 1. A logistic weaning curve and its three milestones, with the default 5% threshold.

The routes through the software

Tooth calculation
slices of one tooth
→ position to age (§3)
→ quality control (§2)
→ curve models (§4)
→ uncertainty, averaging (§4)
→ one animal’s weaning age

Bone calculation
many young animals + adult females
→ quality control (§2)
→ diet and bone-turnover model (§5)
→ Bayesian fit (§5)
→ a site’s weaning age

Enamel (cattle)
$\delta^{18}O$ and $\delta^{13}C$ up one molar
→ yearly $\delta^{18}O$ cycle (§6)
→ season-of-birth index
→ $\delta^{13}C$ rise: rumen development (§6)

Then
compare sites, species or periods (§7)

Before sampling
plan how many samples are needed (§8)

The app has the same order: **Data → Settings → Results → Compare**, plus **Plan sampling**. The Data tab switches between the tooth, bone and enamel calculations.

2. Data and quality control

Input. One CSV or Excel row per tooth slice, or per animal in bone mode; samples can also be typed in by hand. Column names are matched loosely (`Taxon` for `species`, `δ15N` for `d15N` and so on). Tooth data need `specimen_id`, `site`, `species`, `tooth`, `position_mm` (distance from the cusp apex, the tip of the crown, to the middle of the slice) and `d15N`. Bone data need `specimen_id`, `site`, `species` and `d15N`, plus an age at death (`age_months`, or `age_min_months` and `age_max_months`) for each young animal and `age_class = adult` and `sex` for adults. Every row is checked as it is read (`ingest.py`, `schema.py`); rows that fail are listed with the reason, and the rest are kept.

Collagen quality control (`qc.py`). Collagen that has decayed or been contaminated gives misleading isotope values. A sample is flagged when its atomic C:N ratio, %C, %N or collagen yield is outside set limits, and flagged samples are plotted but left out of the fit. Two published sets of limits are offered:

preset	C:N	other limits	use for
Archaeological	2.9–3.6	%C ≥ 13, %N ≥ 4.8, yield ≥ 1%	archaeological collagen (DeNiro 1985; van Klinken 1999)
Modern mammals	3.00–3.28	as above	modern reference animals (Guiry & Szpak 2020)

If C:N is missing it is computed from %C and %N as (%C / %N) × (14.007 / 12.011).

3. Turning tooth position into age

A slice is measured as a distance along the tooth, but a weaning age needs an age. The **growth table** (`growth.py`, `data/growth_tables.json`) gives the age at which dentine formed at several points along a species' tooth; between the points the age is read off a straight line. Beyond the table, ages are extended along the last segment and the samples are marked as extrapolated.

tooth	ages covered	source
Cattle M1	about −2 to 2.5 months	crown completion from Brown et al. (1960); start age and height are placeholders
Cattle M2	about 1 to 12.5 months	crown start and end from Brown et al. (1960); crown height is a placeholder
Human M1	birth to about 9.5 years	Beaumont et al. (2013), after the London Atlas (AlQahtani et al. 2010); provisional
Pig M1, horse M1	—	placeholders

Because an M1 forms too early to show weaning in cattle, an M2 (or an M1 and M2 from the same animal, joined by `individual_id`) is normally needed. Positions measured from the base of the crown are converted using `crown_height_mm`.

Calibration (`calibrate.py`). Labs with known-age reference teeth can build their own table from (position, age) pairs with a straight-line, quadratic or isotonic (any increasing shape) fit. The scatter around the fit becomes the table's **age uncertainty**, which is carried into the weaning-age intervals (§4).

4. The tooth calculation

Curve models

Each specimen's $\delta^{15}N$ series is fitted with five curves (`models.py`). All parametric curves share an adult level `post` and a nursing rise `drop`, and differ only in the shape of the fall:

model	shape	milestones
Logistic	smooth S-shaped fall	from the centre and width, at 5/50/95% of the fall
Piecewise linear	flat, straight fall, flat ("broken stick")	the two break points and their midpoint
Exponential	flat, then fast fall that slows	start, start + τ·ln 2, start + τ·ln 20
Two-phase	two S-shaped steps with a plateau between	first step's onset, half the total fall, second step's completion
Elbow (WEAN-style)	4th-degree polynomial	the "elbow" after the $\delta^{15}N$ peak (completion only)

The logistic and piecewise curves are standard descriptions of a gradual and an abrupt transition. The exponential describes a diet change that is fast at first. The two-phase curve describes staged weaning: supplementary feeding while suckling continues, then final weaning, as expected where people took part of the

milk (Balasse & Tresset 2002). The elbow model re-implements WEAN (Ganiatsou et al. 2023), so results can be compared with that tool; by default both axes are rescaled to 0–1 before the elbow is found, so the answer does not depend on units.

Curves are fitted by least squares with SciPy (`curve_fit`, bounded trust-region method) from 10–20 starting points, keeping the best fit. If the lab's analytical precision is given, it is used as the measurement error of each point.

When no age is reported

A curve can be fitted to any series, including a flat and noisy one, so WeanAge refuses to read a weaning age from it when:

- the fitted fall is smaller than the minimum (default 1‰): **no clear weaning signal**;
- the fall is centred before birth, which cannot be weaning;
- for the elbow model, $\delta^{15}N$ never falls after its peak, or the elbow's 95% interval covers most of the sampled ages.

The curve is still drawn so the data can be inspected. Falls larger than one food-chain step (default 5‰) are flagged, because they suggest another diet change or mixed-up samples.

Ages beyond the samples. A curve can be extended past the last slice, but its shape there is not measured. An age later than the last sample is shown only as **"after X months"** (or "before X months"), without an interval, and is left out of group comparisons.

Uncertainty

Each milestone has a 95% interval. By default, 1000 sets of curve parameters are drawn from the fit's covariance and the milestones recomputed for each; a residual bootstrap (Efron & Tibshirani 1993) can be chosen instead. The growth table's age uncertainty is added as one shared random shift per draw, because an error in the age model moves every slice of a tooth together. The interval is the middle 95% of the draws.

Choosing and averaging models

Models are compared by AICc, Akaike's information criterion corrected for small samples (Hurvich & Tsai 1989). The best-supported model has the lowest AICc. **Akaike weights** turn the AICc differences into shares of support that add up to 1, and the **model average** weights each model's age by its share (Burnham & Anderson 2002). Its interval mixes the models' random draws in the same proportions. An average is given only when the models that report an age hold at least half of the support. The headline result is the best model's age, with the model average beside it (Figure 2).

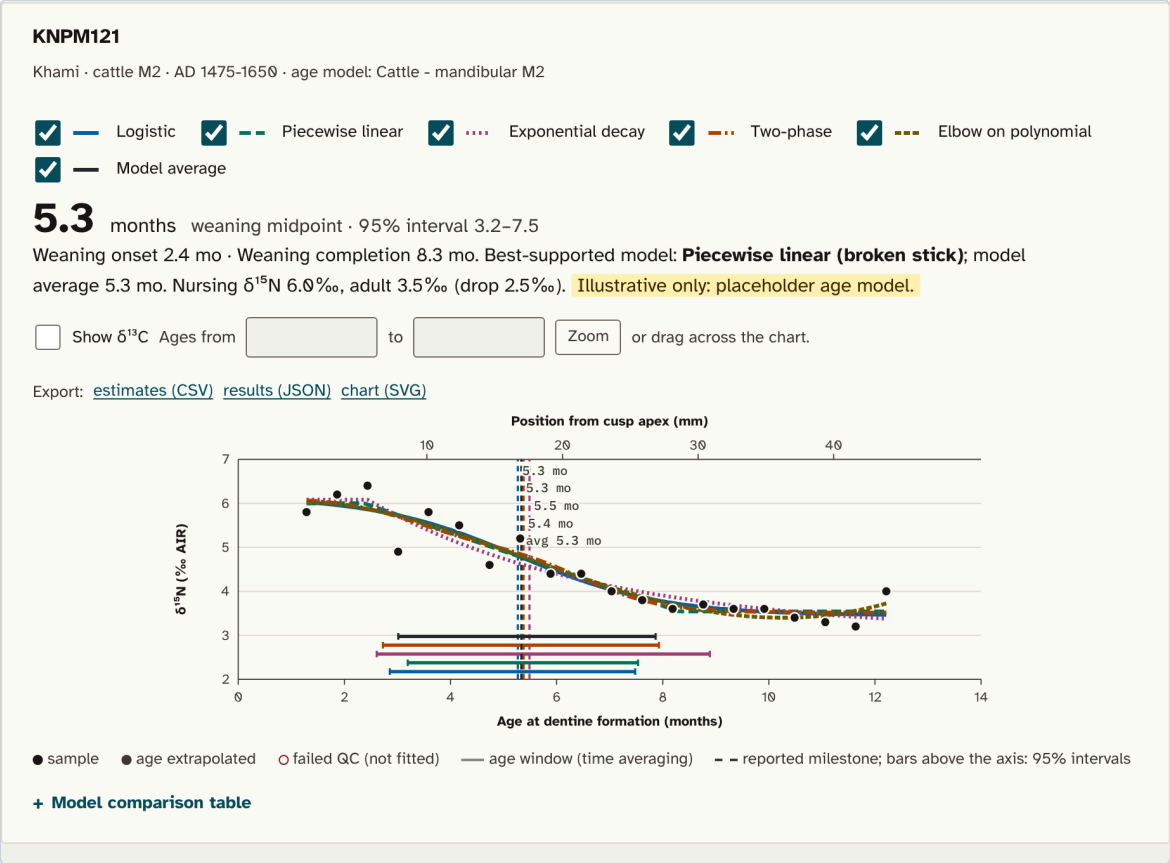


Figure 2. Tooth results for a Khami cattle M2 (real data from IsoArcH; slice positions assumed). The five model curves, their midpoints (dashed lines) and 95% intervals (bars above the age axis).

Corrections that can be switched on

- **Time averaging** (after Tsutaya 2020). A slice mixes dentine formed over several weeks: it has a height, and growth layers run at an angle through its depth. WeanAge covers each slice with an 8 × 8 grid over height and depth, converts each point to an age, and fits the sample as the weighted average of the curve over the grid, with less weight towards the pulp, where there is less dentine. The fitted curve then describes the underlying diet change, which is sharper than the blurred measurements.
- **Seasonal term.** A yearly wave, a·sin(2πt/12) + b·cos(2πt/12), is added to every parametric curve, so a pasture or fodder cycle is not mistaken for weaning. Prehistoric cattle calved seasonally (Balasse et al. 2021), so diet can change with the seasons in the first year.
- **Adult-female baseline** (after WARN; Tsutaya & Yoneda 2013). If the $\delta^{15}N$ of adult females from the same site is entered, results also give the nursing enrichment above them, and the adult plateau can be held within ±2 SD of the female mean.

Warnings

Warnings appear with each result. Besides those above, WeanAge flags:

- a profile that rises with age (Spearman $\rho > 0.3$, $p < 0.05$), since stress and starvation can raise $\delta^{15}N$ (Beaumont & Montgomery 2016);
- for cattle, $\delta^{13}C$ falling by 0.5‰ or more across weaning, because in known-age calves $\delta^{13}C$ rose as $\delta^{15}N$ fell (Balasse et al. 2001), so a fall suggests a different diet change;
- a yearly cycle in the curve's residuals or in $\delta^{13}C$ (an F-test for a sine wave, $p < 0.05$ and at least 0.2‰ or 0.3‰), which may be seasonal diet rather than weaning.

5. The bone calculation

The bone calculation follows WARN (Tsutaya & Yoneda 2013), which was built for human skeletons, and adapts it for animals (`bone.py`). Each site × species (× period) group is fitted on its own. It combines three parts.

Diet. Collagen made at age *t* has $\delta^{15}N = (1 - p)(F + E) + p \cdot W$, after Millard (2000). F is the mothers' mean (adult females), E the nursing enrichment, W the $\delta^{15}N$ of collagen made only from weaning food, and p the share of non-milk food. p is 0 before weaning starts (t1), 1 after it ends (t2), and rises between them; WARN's default shape, $((t - t1)/(t2 - t1))^2$, is used, and linear or reverse shapes can be chosen.

Bone turnover. At birth, bone equals the mothers' value plus a newborn offset b. Every quarter of a month a share of the bone is rebuilt from the current diet, so bone $\delta^{15}N$ follows the diet with a delay. The shares come from turnover tables (`data/bone_turnover.json`). For humans these are WARN's rib collagen values (its Table 1), read as the share replaced per year and converted to monthly shares; this reproduces WARN's 96.6% replaced by 31 weeks. No turnover rates are published for livestock; the cattle, sheep, goat and pig tables are placeholders anchored to young fur seals, whose bone collagen is fully renewed in about 8–10 months (Newsome et al. 2006).

Fit. t1, t2, E, W, F, b and the scatter between animals are estimated together, and each animal's age at death is allowed to vary within its range. WARN fits with approximate Bayesian computation (Sisson et al. 2007); WeanAge writes down the likelihood and samples it with a Markov chain Monte Carlo sampler (Metropolis steps plus differential-evolution moves between 12 chains; ter Braak 2006), checking convergence with $\hat{R}$ (Vehtari et al. 2021). Priors follow WARN where it applies (W ~ N(F, 3), scatter ~ |N(0, 1)|; human E ~ N(1.9, 0.9) from Waters-Rist & Katzenberg 2010). For livestock, E ~ N(2.5–3.0, 1.0) and b ~ N(1.0, 1.0) are placeholders drawn from animal studies (Jenkins et al. 2001; Díez-Canseco & Tornero 2026); t1 and t2 have flat priors up to a user-set latest end of weaning, to avoid importing human weaning ages. t2 is reported as the end of suckling (§1).

Checks. Reynard & Tuross (2015) found that a raised $\delta^{15}N$ in the very young is common but weaning ages are hard to extract, often because samples are small. Before fitting, the youngest third of the young animals are compared with the mothers (one-sided Welch t-test); if they are not clearly higher, a warning says the data may hold no nursing signal. Warnings also appear for fewer than 10 young animals, intervals that only repeat the prior, and $\hat{R}$ above 1.1. As Tsutaya & Yoneda stress, the model always fits something, so look at the plot first.

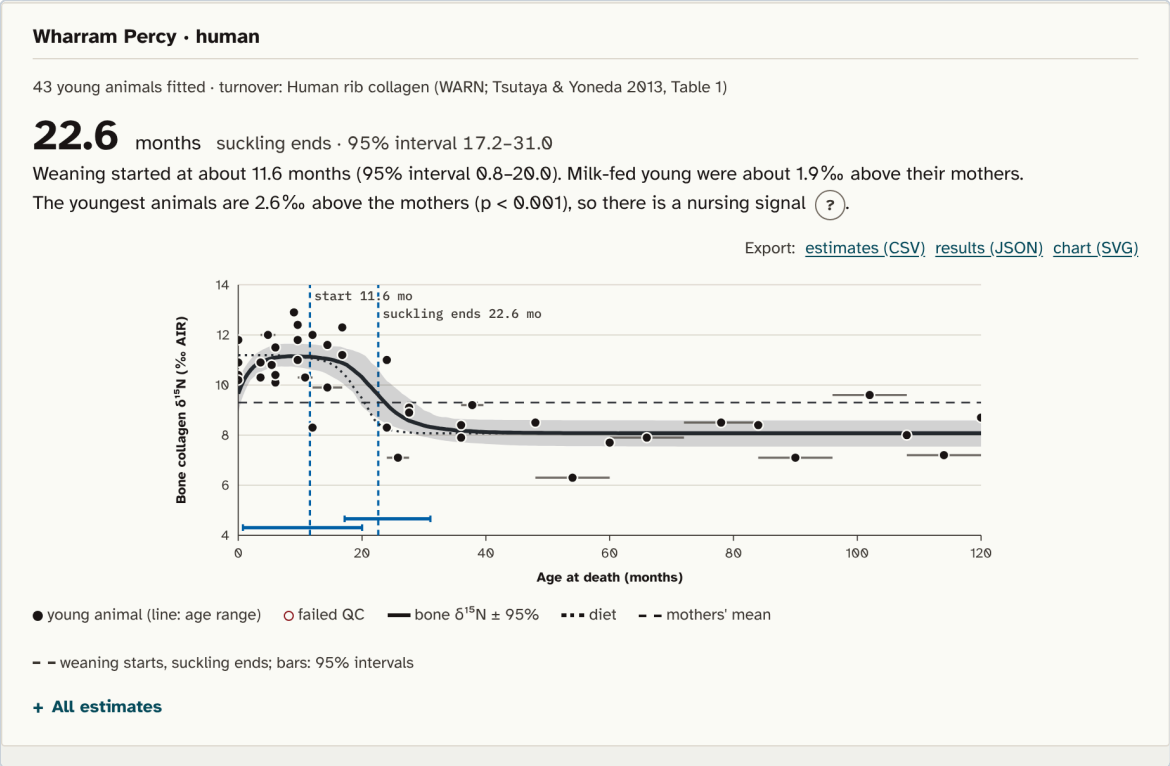


Figure 3. Bone results for medieval Wharram Percy (Richards et al. 2002, via CIMA). Points are children's rib collagen with their age ranges; the solid line is the modelled bone $\delta^{15}N$, the dotted line the diet. Suckling ends at 22.6 months (95% interval 17–31), matching the authors' estimate of about two years.

6. Season of birth and rumen development (enamel)

This section uses tooth enamel rather than collagen, and is set up for cattle (`enamel.py`). Enamel of a molar forms from the cusp tip down to the enamel-root junction (ERJ) over several months, so samples drilled up the crown are a time series. Each sample's position is its distance from the ERJ (`erj_distance_mm`); values are enamel carbonate $\delta^{18}O$ and $\delta^{13}C$ (‰ VPDB).

Season of birth ($\delta^{18}O$). Enamel $\delta^{18}O$ follows drinking water, which is higher in summer. Each tooth's $\delta^{18}O$ series is fitted with a cosine, $\delta^{18}O(x) = M + A \cdot \cos(2\pi(x - x_0)/X)$, after Balasse et al. (2012): X is the crown length formed in a year and x_0 the position of the $\delta^{18}O$ maximum. The **birth index** x_0/X (0–1) places that summer on the tooth. Teeth of the same type start forming at about the same age, so differences in x_0/X between animals reflect differences in season of birth (cattle: Towers et al. 2014; Balasse et al. 2021). It is a relative index, not a date. For each site, the spread of the indices × 12 approximates how many months calving was spread over. No index is given when the $\delta^{18}O$ cycle is small (default under 0.5‰) or a cosine does not fit clearly better than a flat line.

Rumen development ($\delta^{13}C$). A suckling calf's rumen is not yet working. As it develops, fermentation enriches the carbon that reaches the enamel in ^{13}C (Passey et al. 2005; Cerling & Harris 1999), so first molar enamel $\delta^{13}C$ rises (Towers et al. 2014). Because plant $\delta^{13}C$ also changes with the seasons, $\delta^{13}C$ is fitted twice: with a yearly wave of the period found from $\delta^{18}O$, and with the same wave plus a rising step. The step is reported as rumen development only when it fits clearly better (AICc) and rises by at least 0.5‰; its position on the crown has a 95% interval. Rises above 3‰ are flagged, since rumen development alone gives about 1–2‰ and a change of diet (for example to C4 plants) may be involved.

Enamel takes months to mineralise at any one position (Zazzo et al. 2005), so both signals are smoothed: the real $\delta^{18}O$ cycle was larger and the $\delta^{13}C$ change sharper than recorded. Positions are not converted into ages, because enamel formation tables for cattle are not yet available.

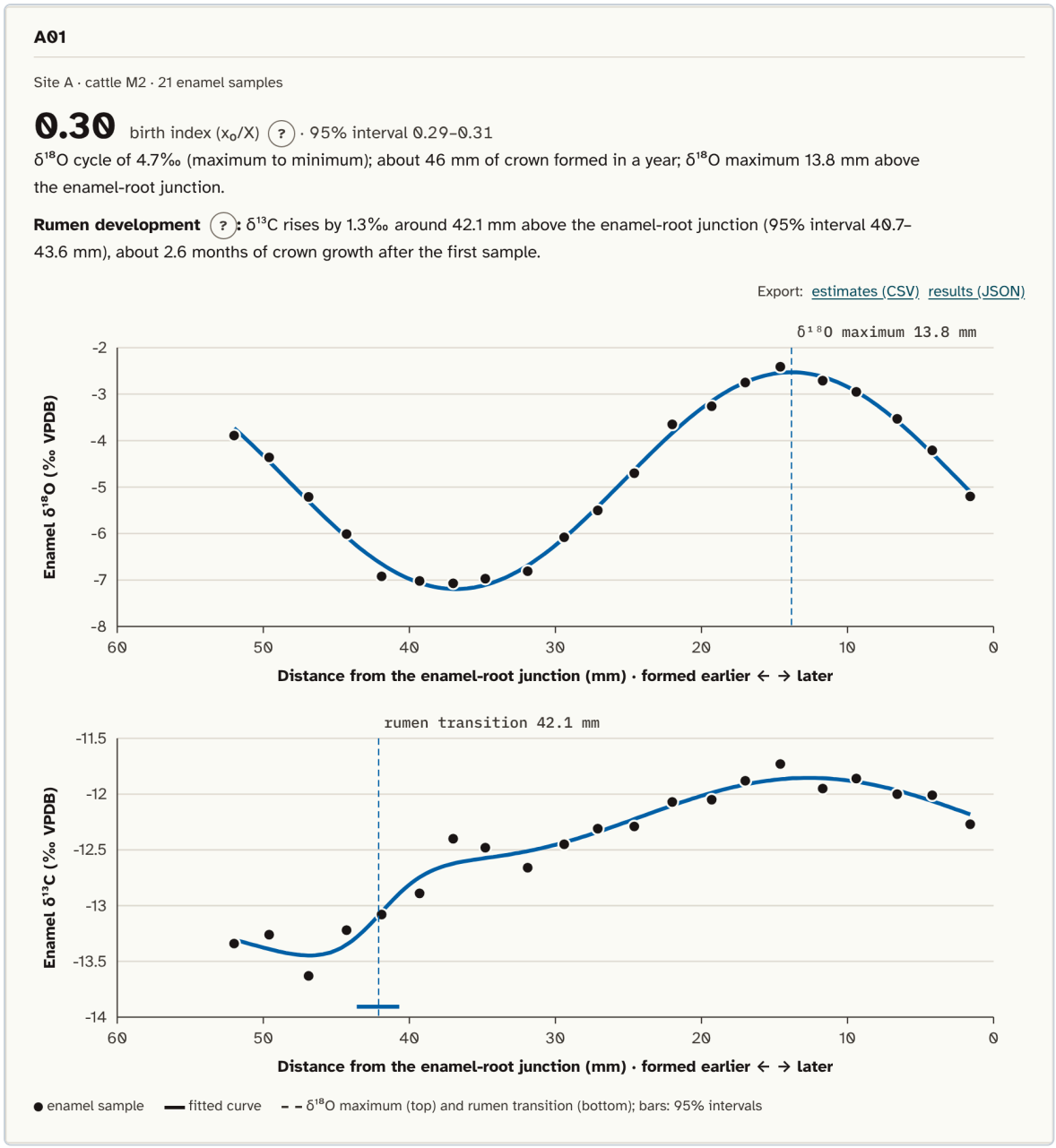


Figure 4. Enamel results for a synthetic cattle M2 (simulated data). Top: $\delta^{18}O$ along the crown with the fitted cosine and its maximum. Bottom: $\delta^{13}C$ with the fitted wave and rumen step, and the transition with its 95% interval.

7. Comparing groups

The Compare tab groups tooth estimates by site, species and/or period (`aggregate.py`, `inference.py`, `joint.py`).

- **Summary statistics**: n, mean, SD, median, range and a 95% confidence interval.
- **Random-effects means** (DerSimonian & Laird 1986): each group's mean age, allowing for real differences between animals (τ) as well as each estimate's own uncertainty, taken from its 95% interval. A Q-test asks whether groups differ at all; pairwise tests use Holm's (1979) correction for multiple comparisons; a permutation test (5000 shuffles of the group labels) checks the result without assuming normal errors.
- **Hierarchical Bayesian model** (two-stage): each specimen's estimate is treated as a noisy measurement of its true age, and true ages vary around a group mean. The between-animal SD has a half-Cauchy prior (Gelman 2006). It gives the probability that one group weaned earlier than another, and partially pooled specimen estimates.

- **One-stage model**: the logistic curves of all specimens and the group comparison are estimated together from the raw $\delta^{15}N$ values, so uncertainty in each curve's shape and each tooth's age model is carried through. It uses the logistic curve only.

Validation. If known weaning ages (`known_weaning_age_months`) or published estimates are included, the Compare tab reports each model's bias, mean absolute error, root-mean-square error, correlation and 95% interval coverage.

8. Planning a study

Both planners answer "would this sampling design work?" before teeth are cut or bones sampled.

Teeth (`simulate.py`). The planner invents teeth with known weaning curves, "cuts" them as the lab would (slice spacing and height, angled growth layers, measurement noise, lost samples, age-model error), runs the full analysis, and scores each model's completion age against the truth. Trying one setting at several values (for example 0.5, 1 and 2 mm slices) with the same random seed shows which protocol is accurate enough. Simulating curves of other shapes than the logistic shows how each model copes when its shape is wrong.

Bones (`bone_planner.py`). The planner invents sites with a known end of suckling and samples them as planned: the number of young animals, how their ages are spread, how precisely they are aged, and the number of adult females. It runs the bone calculation on each site and reports the error, the width and coverage of the 95% interval, and how often the nursing signal is found. Because the data are simulated with the model that is fitted, it tests the design, not the model; with placeholder turnover it shows relative, not absolute, precision.

9. Using the app

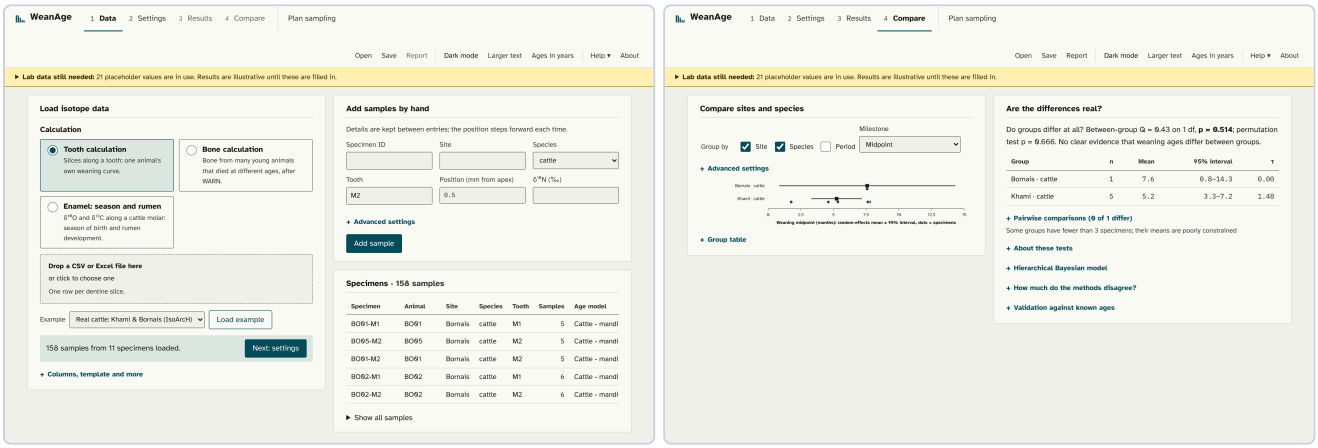


Figure 5. The Data tab with the cattle example loaded (left) and the Compare tab (right).

1. **Data.** Choose the tooth, bone or enamel calculation, then drop a file, load an example or type samples in. Problems are listed with their row numbers.
2. **Settings.** The main choices are shown; less used ones are under *Advanced settings*. The defaults suit a first run.
3. **Results.** Pick a specimen or group on the left. The headline gives the reported age and its 95% interval; the chart, the model table and the warnings explain it. Results can be exported as CSV, JSON or SVG, and the whole run as an HTML report.
4. **Compare** (tooth data) tests differences between groups.
5. **Plan sampling** runs the planners.

Save writes the data, settings and results to a project file; work is also kept in the browser. *Ages in years* switches every result to years. Every **?** opens a short explanation. Examples from IsoArcH (cattle teeth from Khami and Bornais) and CIMA (Cocozza et al. 2022; human bone from Wharram Percy, York Fishergate and Newark Bay, and

human teeth from Mikulčice) are included; their slice positions or tooth age models are partly assumed, as noted when they load.

10. Limits and placeholders

Values that still need laboratory data are marked **placeholder** in the app and listed in the yellow banner at the top. Until they are replaced, results are illustrative. The most important are:

- crown heights and formation ages along the tooth for cattle, pig and horse, and a checked human M1 table;
- bone collagen turnover rates and nursing-enrichment priors for livestock;
- collagen quality-control limits and the minimum weaning drop for each laboratory.

Other limits: growth layers are treated as straight and the tooth as round in cross-section; the one-stage model uses only the logistic curve; and the bone calculation is not yet part of the Compare tab or the report. Animals that died young are not a random sample of the herd (for example culled surplus calves in dairying), just as children who died are not a random sample of the living, which affects which ages are represented.

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Example data: IsoArcH (doi:10.48530/isoarch.2021.014, CC BY 4.0; Dyvart et al. 2018; Griffith & Mulville 2020) and CIMA (doi:10.48493/s9nf-1q80, CC BY-SA). WeanAge re-implements the methods of WARN, MDSS and WEAN from their publications; no code was copied.